

PBO Research Roadmap v1.0

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2026-07-16

Pattern-Based Ontology Research Roadmap v1.0

Rebranded from Autaxys Master Plan v3.0

Author: Rowan Brad Quni-Gudzin **Date:** 2026-07-16 (v1.0; reformulation of Autaxys Master Plan v3.0 of 2026-07-15) **Canonical Framework:** [Pattern-Based Ontology v1.0](#) **Prior DOI:** [10.5281/zenodo.21369313](#) (Autaxys Master Plan v3.0) **License:** QNFO Unified License Agreement (QNFO-ULA)

Reformulation Note

This document is the rebranded successor to the Autaxys Master Plan v3.0 (2026-07-15).

The Autaxys research program has been systematically reformulated as a **Pattern-Based Ontology** grounded in Spencer-Brown’s *Laws of Form*. The transition is documented in full in [Pattern-Based Ontology v1.0](#) (2026-07-16). Key changes:

Aspect	Master Plan v3.0 (This document’s predecessor)	Roadmap v1.0 (This document)
Foundation	“Autaxys” as frequency-centric process ontology	Pattern-Based Ontology — recursive distinction as primitive operation
Primitive operation	5 dynamics + 5 meta-logical principles	1 mark + 2 reduction rules (Calling, Crossing)
Core identity	$m = \omega$ (mass-frequency)	$m = \omega$ (pattern iteration rate — same math, clearer ground)
Geometry	Implicit (relational)	Explicit (Bruhat-Tits ultrametric tree)
Ontological commitment	Process ontology (v3.0 was interpretation-neutral)	Structure-neutral — compatible with both process and static views
Name origin	Greek: <i>auto</i> (self) + <i>taxis</i> (arrangement)	Descriptive: patterns as fundamental ontological category

What is preserved: The entire research program structure — research pillars, project templates, methodology, publication registry, and strategic roadmap — is carried forward. The

reformulation affects *framing*, not *substance*. The research tracks identified in Master Plan v3.0 remain active under new names and cross-references.

0.0 Framework Foundation

The Pattern-Based Ontology (PBO) posits that all of reality consists of **patterns** generated by the recursive application of a single primitive operation: **drawing a distinction** (Spencer-Brown, 1969). From this operation — the mark # and enclosure [] — and two reduction rules (Calling: ## → #; Crossing: [#] →), the entire universe of structure emerges.

The Bridge Equation survives the reformulation intact:

$$\hbar\omega_C = m_0c^2 \quad \Rightarrow \quad m = \omega \text{ (natural units)}$$

Under PBO, mass is not a “quantity of substance” but the **iteration rate of a stable pattern** — the frequency at which a pattern’s internal distinctions reiterate to maintain its boundary against the surrounding pattern network.

Certainty: All claims in this roadmap carry the default label [my conjecture] unless labeled otherwise. [established] claims reference textbook-consensus physics. The core thesis — that all is pattern — remains a [PHILOSOPHY] position, though specific physical predictions (CMB oscillations, spacetime granularity, particle normal forms) are [speculative, not yet falsifiable].

Map/Territory Distinction: This roadmap describes a research program — a map. Whether the territory corresponds to the map is an open question. The mathematical structure ($m = \omega$, the ultrametric tree, the STC) is compatible with multiple ontologies.

1.0 Research Pillars (Rebranded from Master Plan §5.0)

The PBO research program is organized into six pillars. Each pillar maps directly to a previously defined Autaxys research track, now grounded in the distinction calculus.

Pillar	Autaxys Predecessor	Description	Status
P1: STC Formalization	Formalisms / Relational Calculus	Complete mapping from Syntactic Token Calculus normal forms to Standard Model particles	Active
P2: Cosmological Signatures	Dark Sector / CMB	Log-periodic CMB oscillations, spacetime granularity, dark energy as distinction-tree expansion	Pending
P3: Ultrametric Quantum Computing	Kepler Program / Adelic QC	Passive geometric fault tolerance via Bruhat-Tits tree encoding	Active

Pillar	Autaxys Predecessor	Description	Status
P4: Gravity Reformulation	Relational Graph Gravity	Derive Einstein field equations from pattern interaction graph	Conceptual
P5: Consciousness as Pattern	S / Emergence	Self-referential pattern — the distinction that includes the distinguisher	Conceptual
P6: Cross-Disciplinary Validation	ARM / Applied Research	Test pattern hierarchy predictions in biology, neuroscience, sociology	Future

1.1 Methodology (Rebranded from Master Plan §7.0)

The methodology remains: strict constructionist desk research → mathematical formalization → computational modeling → prediction generation → empirical test. The key methodological principle is **derivation from first principles**: every claim must trace back to the mark and the two reduction rules, or be explicitly flagged as external.

1.2 Publication Registry (Updated)

Date	Publication	DOI	Role in PBO
2026-07-16	Pattern-Based Ontology v1.0	10.5281/zenodo.2138957	Core conceptual framework document
2026-07-16	42 Theses v3.0	10.5281/zenodo.2138947	42 theses with certainty calibration
2026-07-15	Pattern Hierarchy v3.0	pending	Reformulated taxonomy (was: Autaxic Table)
2026-07-15	PBO Research Roadmap v1.0	pending	This document (was: Autaxys Master Plan)
2026-07-14	Mass-Frequency Identity v3.2.0	10.5281/zenodo.2138949	2138949 equation, ontological neutrality
2026-07-04	The Resonant Universe v2.0.1	10.5281/zenodo.2119186	Formal treatise (title retains “Autaxys” — update pending)
2026-07-15	Ontological Status Addendum v1.0	10.5281/zenodo.2136930	2136930 information bridge document
2026-04-15	Quantum Laws of Form v3.0	10.5281/zenodo.1957801	1957801 formalization, particle taxonomy

2.0 Strategic Timeline

Era	Timeline	Focus
Phase 2 (Now)	2026 Q3-Q4	STC → Standard Model mapping (P1); CMB prediction (P2); Ultrametric QC prototype (P3)
Phase 3	2027	Gravity reformulation (P4); Consciousness model formalization (P5)
Phase 4	2028+	Cross-disciplinary validation (P6); Empirical tests of falsifiable predictions

2.1 Immediate Priorities

1. **Complete STC particle taxonomy** (mapping all Standard Model particles to normal forms) — blocking P1
2. **Derive CMB log-periodic oscillation parameters** — first concrete, falsifiable prediction from P2
3. **Prototype ultrametric quantum gate** — proof-of-concept for P3
4. **Update remaining Autaxys-branded Zenodo records** — rebrand “Autaxys” → “Pattern-Based Ontology” where applicable

3.0 Remaining Autaxys Material — Rebranding Survey

The following Zenodo records in the “Autaxys and Autology” community carry “Autaxys” or “Autaxic” branding and are candidates for rebranding. Records are grouped by priority.

TIER 1 — Direct Rebranding (content is active, name should change)

#	Record	DOI	Action
1	The Resonant Universe: Autaxys , a Formal Treatise v2.0.1	10.5281/zenodo.21191864	2024 version: update title to “Pattern-Based Ontology: A Formal Treatise”
2	Autaxys and Autology: Frequency-Centric Process Ontology	10.5281/zenodo.15782973	2023 version: “Pattern-Based Ontology and Autology”
3	The Autaxic Trilemma: A Theory of Generative Reality	10.5281/zenodo.15715865	2022 version: “The Pattern Trilemma”
4	A (Very) Brief Guide to Autaxys	10.5281/zenodo.15710936	2022 version: “A Brief Guide to Pattern-Based Ontology”
5	AUTX D-P6.2-2/4: Autaxys Formalisms	10.5281/zenodo.15682800	2020 version: “PBO Formalisms / Formal Toolkit”
6	Autaxys and its Generative Engine	10.5281/zenodo.15557480	2020 version: “Pattern-Based Ontology and the Distinction Engine”

TIER 2 — Add Reformulation Note (superseded or historical)

#	Record	DOI	Action
7	Autaxys Phase 1 Milestone Evaluation	10.5281/zenodo.15714046	Footnote: “This evaluation documents the pre-reformulation framework. See PBO v1.0 for the reformed version.”
8	Autaxic Table of Patterns v1 (superseded)	10.5281/zenodo.15628365	Footnote: superseded by Pattern Hierarchy v3.0
9	Autaxys R&D Master Plan v2 (superseded)	10.5281/zenodo.15617573	Footnote: superseded by PBO Research Roadmap v1.0

TIER 3 — No Change Needed (archives, non-Autaxys-named)

#	Record	DOI	Action
10-18	Archives (FID, PBRF, IO, Infomatics, Autaxys archive)	various	No change — historical archives
19-20	Natural Units, Critical Examination	various	No change — not Autaxys-branded
21	Mass-Frequency Identity v3.2.0	10.5281/zenodo.21360549	Already rebranded (stripped “Autaxys” in v3.2)

GitHub Repos with Autaxys Material

Repo	Status	Action
rwnq8/autaxys	Archived (Mar 2026)	Add README note pointing to PBO repos. Cannot modify archived content.
rwnq8/autologos	Archived (Jul 2025)	Historical — no change needed
rwnq8/primordial-mark	Active	Already PBO-aligned (distinction as generative engine)
rwnq8/qlof-primer	Active	Already PBO-aligned (distinction-first pedagogy)
rwnq8/quantum-laws-of-physics	Active (implied)	Already PBO-aligned (STC formalism)
rwnq8/qnfo-skills	Active	Update publication-publisher skill to reference PBO pattern

4.0 Version History

Version	Date	Changes
v1.0	2026-07-16	Rebranded from Autaxys Master Plan v3.0. Added Reformulation Note. Replaced “Autaxys” terminology with “Pattern-Based Ontology.” Added §0.0 Framework Foundation grounding in Laws of Form distinction calculus. Replaced 5+5 generative engine with 1+2 primitive operations. Added rebranding survey (Tier 1–3). Cross-referenced PBO v1.0 as canonical.
v3.0 (predecessor)	2026-07-15	Added ontological status, Map/Territory, framework limitations, external literature positioning
v2.0	2025-06-08	Original segmented Master Plan

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This document was prepared by a QNFO Research Agent (DeepSeek v4 Pro). It rebrands and supersedes the Autaxys Master Plan v3.0 (DOI: 10.5281/zenodo.21369313) as part of the systematic reformulation of the Autaxys research program into a Pattern-Based Ontology grounded in Laws of Form. Canonical framework: [PBO v1.0](#).